CMSE 202 Project Final Report

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Introduction

In this project, we aimed to answer a simple yet interesting question: what makes a song popular on Spotify? With millions of users and a huge catalog of music, Spotify tracks many audio features and assigns each song a popularity score between 0 and 100. We wanted to know which features—danceability, energy, tempo, or loudness, for example—go into making a song more popular.

Methodology

We used a Kaggle dataset that included over 160,000 tracks from the year 2006 to 2020, with each track including extensive information about various music-related aspects. We first imported required Python libraries like Pandas, Seaborn, Matplotlib, and Scikit-learn, then filtered the data to include songs from the year 2006 and beyond. We inspected the data structure and cleaned the data by removing any missing values so that we could ensure our analysis would be valid.

Data Analysis

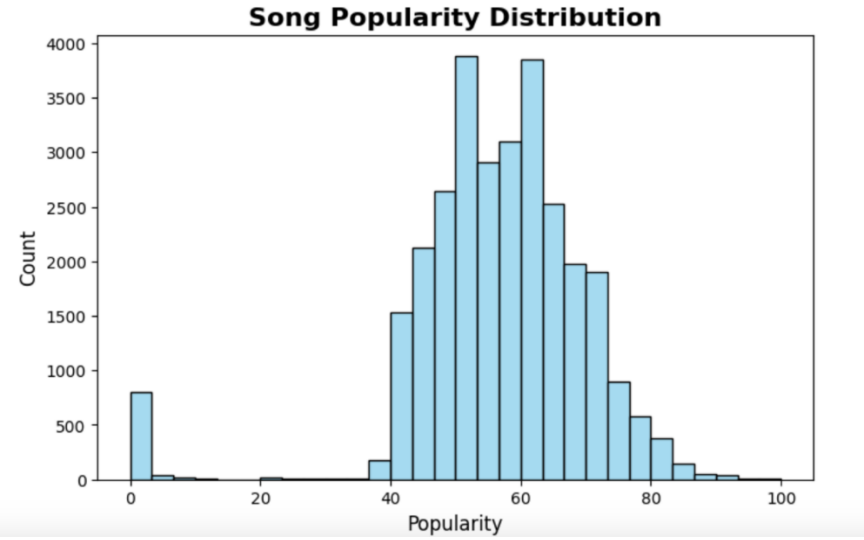


Figure 1. *Song Popularity Distribution.*
Histogram showing the distribution of song popularity scores from 0–100. Most songs fall between 40 and 70 in popularity, with far fewer songs at the extreme ends.

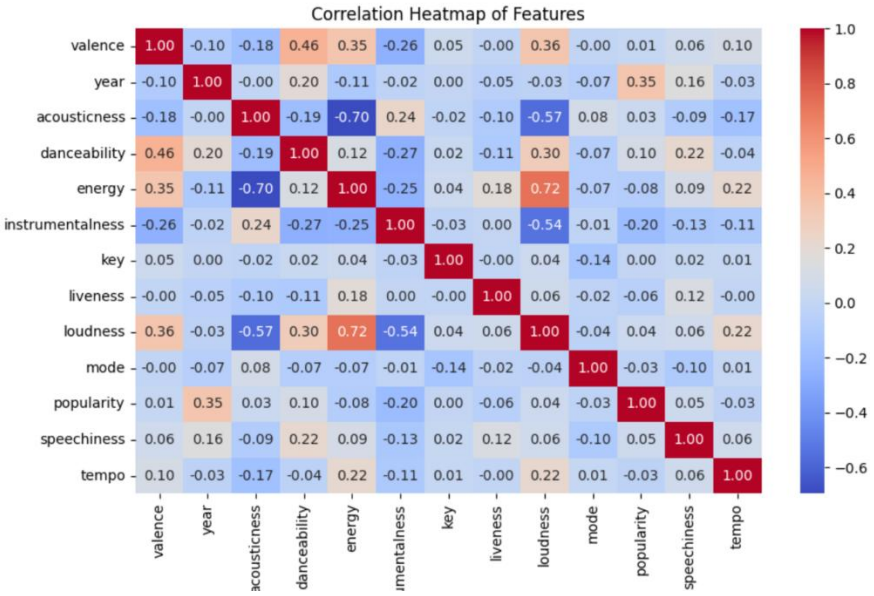
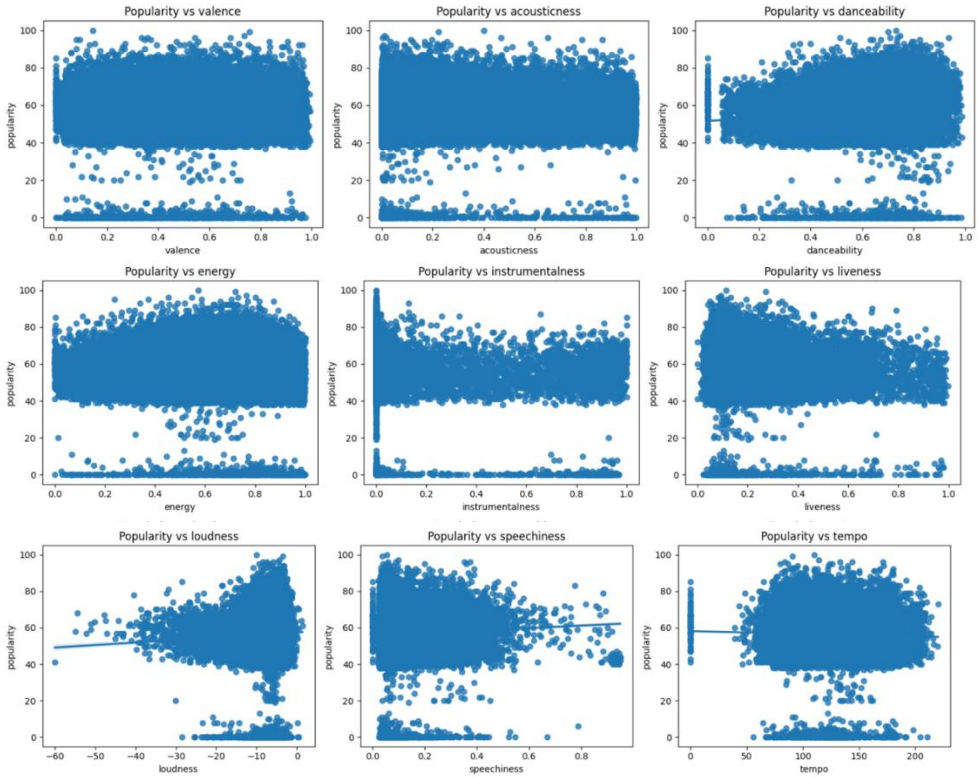


Figure 2. *Correlation Heatmap of Features.*
Heatmap showing correlations between various audio features and popularity. Danceability and energy show small positive correlations, while acousticness has a small negative correlation.



Figures 3–11. *Scatter Plots of Audio Features vs. Popularity.*
Scatter plots comparing each audio feature (valence, acousticness, danceability, energy, instrumentalness, liveness, speechiness, tempo, and loudness) to popularity. Most features show weak or no correlation with popularity, with danceability having the most noticeable positive trend.

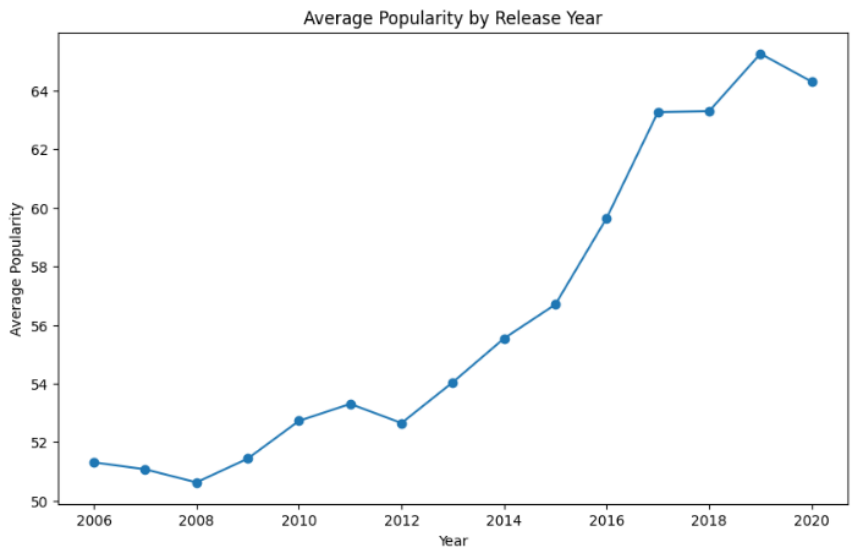


Figure 12. *Average Popularity by Release Year.*
This line plot shows how average popularity has changed from 2006 to 2020. Popularity stayed stable between 2006–2013, rose steadily from 2014 onward, and peaked in 2019 before a slight dip in 2020.

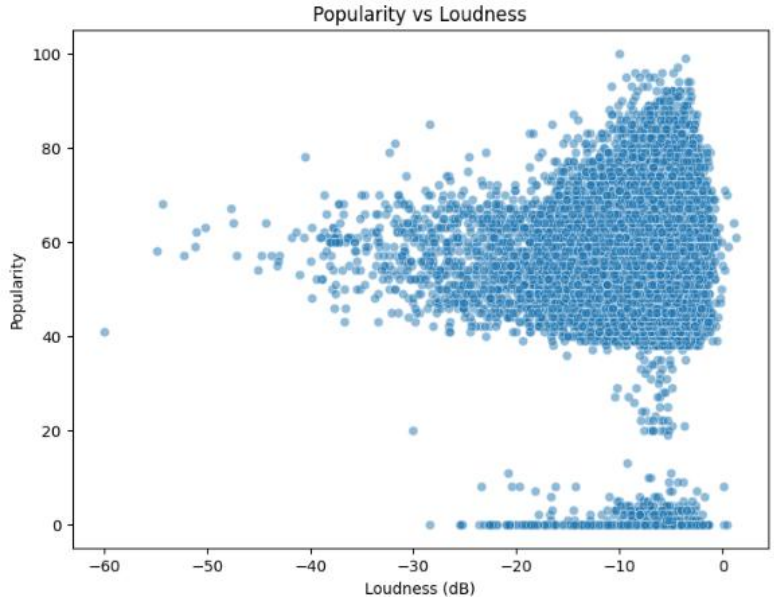


Figure 13. *Popularity vs. Loudness.*
This scatter plot compares loudness (in decibels) with popularity scores. Popular songs appear across all loudness levels, with no clear trend, suggesting loudness alone does not determine popularity.

Results

From our analysis, we found that newer songs are generally more popular, especially after 2014, with the highest average popularity in 2019 (Figure 12). Our linear regression model showed that energy (33.15) and danceability (30.63) had the biggest positive effects on popularity. Speechiness and loudness had smaller positive effects, while instrumentalness and acousticness had weak negative effects, meaning songs that were more acoustic or instrumental-heavy tended to be slightly less popular. Looking at the scatter plots (Figures 3–11), we saw that while energetic and danceable songs often scored higher, there were plenty of popular songs that didn't fit those patterns, showing that these features don't fully explain popularity.

Challenges Faced

We initially planned to use the Spotify API, but it caused authentication errors and even crashed our JupyterHub kernel. With limited time and experience, we decided to drop the idea and focus on the Kaggle dataset. Cleaning the large dataset was time-consuming, and many audio features showed weak or unclear relationships with popularity. We also had occasional GitHub merge conflicts while collaborating.

Conclusion

Overall, features like danceability, energy, and release year do seem to matter, but they don't tell the whole story. Loudness didn't show a strong overall effect, even though some popular songs were on the louder side. Our regression model had an MSE of 185.63, which means audio features alone can't predict popularity very accurately. Things like artist popularity, marketing, playlist placement, and cultural trends are probably much bigger factors. If we had more time, adding things like genre, artist stats, or playlist ranking could make the predictions more accurate and give a better idea of what actually makes a song popular.